

1. Administrative & Study Identification

Full Study Title: A Long-Term Safety Extension Study of Mavacamten (MYK-461) in Adults With Hypertrophic Cardiomyopathy Who Have Completed the MAVERICK-HCM (MYK-461-006) or EXPLORER-HCM (MYK-461-005) Trials (MAVA-LTE) **Short Title/Acronym:** MAVA-LTE **Posting ID:** 497642017515532565 **Protocol Number:** MYK-461-007 **NCT Number:** NCT03723655 **Trial Status:** ACTIVE_NOT_RECRUITING **Sponsor/Company Name:** Bristol Myers Squibb (formerly MyoKardia Inc.) **Investigational Product:** Mavacamten (CAMZYOS) | **ATC Code:** C01EB24 **Phase of Development:** Phase II/III (Long-Term Extension) **Medical Monitor:** Bristol Myers Squibb Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the long-term safety and tolerability of mavacamten over a 5-year period in adults with symptomatic hypertrophic cardiomyopathy (HCM), including obstructive hypertrophic cardiomyopathy and non-obstructive hypertrophic cardiomyopathy, who completed the EXPLORER-HCM or MAVERICK-HCM trials. **Secondary Objectives:** To evaluate the long-term efficacy of mavacamten, including durable improvements in left ventricular outflow tract (LVOT) gradients, N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels, New York Heart Association (NYHA) functional class, and patient-reported outcomes (e.g., KCCQ score).

2.2 Background & Rationale Mavacamten is a first-in-class cardiac myosin inhibitor that reduces actin-myosin cross-bridge formation, normalizing hypercontractility in HCM patients. While the parent Phase 3 EXPLORER-HCM and Phase 2 MAVERICK-HCM trials demonstrated significant short-term efficacy and safety, data assessing the long-term safety, durability of clinical response, and management of transient reductions in left ventricular ejection fraction (LVEF) under extended treatment are required.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (within cohorts) **Blinding:** Open-label (Dose-blinded during the initial parent-study parallel phase, subsequently open-label) **Randomization:** Parallel assignment based on parent study dosing regimens

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 282 patients **Number of Sites:** Approximately 30 sites from MAVERICK-HCM (US) and approximately 90 sites from EXPLORER-HCM (US, Europe, and Israel), with approximately 30 centers overlapping between MAVERICK and EXPLORER. **Estimated Study Start:** April 2019 **Estimated Primary Completion:** September 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult subjects (≥ 18 years) who have completed the Parent Study (EXPLORER-HCM or MAVERICK-HCM) through to the End of Study (EOS) visit within 90 days of consent. (Participants beyond the 90-day window may be included pending Medical Monitoring approval. Participants who prematurely discontinued from the Parent Study or MAVA-LTE study may be considered for inclusion). 2. Documented resting Left Ventricular Ejection Fraction (LVEF) $\geq 50\%$ by echocardiography core laboratory read of screening TTE at rest. 3. Body weight greater than 45 kg at the Screening Visit, with adequate acoustic windows to enable accurate TTEs. 4. Has safety laboratory parameters (chemistry, hematology, coagulation, and urinalysis) within normal limits (according to the central laboratory reference range). 5. Female participants must not be pregnant or lactating and, if sexually active, must use a highly effective birth control method from Screening through 90 days after the last dose of investigational medicinal product (IMP). Male partners must also use a contraceptive (e.g., barrier, condom, or vasectomy).

4.2 Exclusion Criteria 1. History of syncope, history of sustained ventricular tachyarrhythmia with exercise, resuscitated sudden cardiac arrest, or known history of appropriate implantable cardioverter-defibrillator (ICD) discharge for life-threatening ventricular arrhythmia between Parent Study EOS Visit and Screening Visit. (Note: history of anti-tachycardia pacing (ATP) is allowed). 2. Current treatment with disopyramide or ranolazine within 14 days prior to screening, or planned treatment with these agents during the study. 3. Any ECG abnormality considered by the investigator to pose a risk to participant safety (e.g., second degree atrioventricular block type II), or history of clinically significant malignant disease developed since enrollment in the Parent Study. 4. Has any acute or serious comorbid condition (e.g., major infection or hematologic, renal, metabolic, gastrointestinal, or endocrine dysfunction) that could lead to premature termination or interfere with efficacy and safety assessments. 5. Is unable to comply with the study requirements, including the number of required visits to the clinical site. 6. Has participated in a clinical trial receiving any investigational drug (or device) within 30 days prior to Screening, or 5 times the respective elimination half-life, except for participation in MAVERICK-HCM or EXPLORER-HCM. Prior participation in a non-interventional observational study is allowed.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Mavacamten is initiated at 5 mg once daily. Subsequent targeted dose titration (to 2.5 mg, 5 mg, 10 mg, or 15 mg) is performed based on site-read echocardiographic evaluation of Valsalva LVOT gradient and LVEF. **Route of Administration:** Oral **Duration of Treatment:** Up to 252 weeks (~5 years) of exposure.

5.2 Concomitant & Prohibited Therapy Permitted: Standard background medical therapies for HCM (such as beta-blockers or calcium channel blockers, provided they are not strictly prohibited). **Prohibited:** Disopyramide, ranolazine, and background combination therapy of beta-blockers with non-dihydropyridine calcium channel blockers (verapamil or diltiazem).

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Day -14 to 0	Washout from parent study, Informed Consent, Baseline TTE, Labs
Baseline	Day 0	Initiation of 5 mg mavacamten daily
Titration / Interim	Weeks 4, 8, 12, 24	Site-read TTEs for dose adjustments based on LVOT gradient and LVEF
Maintenance	Every 12 weeks	Routine safety echocardiograms, AEs monitoring, NYHA, and Labs (e.g., Week 156, Week 180)
End of Study	Week 252	Final Safety Follow-up and TTE assessments

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Frequency and severity of treatment-emergent adverse events (TEAEs) and serious adverse events (SAEs). A key safety measure includes the tracking of transient reductions in LVEF $< 50\%$ requiring temporary treatment interruptions. **Measurement Method:** Continuous clinical monitoring, laboratory assessments, and protocol-defined sequential echocardiography throughout the 252-week time frame.

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in resting and Valsalva LVOT gradients. 2. Change from baseline in N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels. 3. Change from baseline in NYHA functional class and patient-reported quality-of-life parameters.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular 12-week site visits. The protocol specifies strict criteria for

temporary study drug interruption if LVEF falls below 50%. **Serious Adverse Events (SAEs):** Evaluated continuously and reported within protocol-defined regulatory timelines. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (IDMC) periodically reviews cumulative safety data, particularly focused on occurrences of heart failure symptoms and LVEF changes.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The Safety Set includes all patients who receive at least one dose of the investigational product in the MAVA-LTE study. **Sample Size Justification:** Not powered for formal hypothesis testing. The sample size ($N = 282$) is based on the number of eligible patients who completed the parent EXPLORER-HCM and MAVERICK-HCM trials and consented to the extension. **Handling of Missing Data:** Summarized primarily using descriptive statistics. Longitudinal continuous efficacy variables (e.g., LVOT gradients, NT-proBNP) are assessed based on observed data at prespecified time points without explicit imputation for dropouts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict echocardiographic core lab standardization and onsite monitoring to ensure protocol adherence regarding dose titrations. **Audit Trail:** Managed via secure Electronic Data Capture (EDC) systems featuring investigator sign-offs and data clarification protocols.

11. Study Identifiers & Governance

Primary ID: MYK-461-007 **Registry Number:** NCT03723655 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Long-Term Safety Extension Study of Mavacamten in Adults Who Have Completed MAVERICK-HCM or EXPLORER-HCM **Official Regulatory Title:** A Long-Term Safety Extension Study of Mavacamten (MYK-461) in Adults With Hypertrophic Cardiomyopathy Who Have Completed the MAVERICK-HCM (MYK-461-006) or EXPLORER-HCM (MYK-461-005) Trials (MAVA-LTE)

12. Clinical Framework (HCM Specific)

Disease Areas / Primary Conditions: Hypertrophic Cardiomyopathy, Obstructive Hypertrophic Cardiomyopathy, and Non-obstructive Hypertrophic Cardiomyopathy. **Ontology Codes:** MeSH: D002312 | SNOMED CT: 233873004 | HPO: HP:0001639 **Diagnosis Threshold:**

Pathologically thickened left ventricular wall in the absence of other loading conditions, with baseline LVEF $\geq 50\%$. **Medical Methodology:** Cardiac Myosin Inhibition **Optimization Target:** Normalization of the LVOT gradient (≤ 30 mmHg) and improvement of symptoms (NYHA functional class) while strictly maintaining LVEF $\geq 50\%$.

13. Study Procedures & Methodology

Management Pathway: Targeted dose-titration driven by echocardiographic response. **Education Protocol (HCP):** Site-level training on strict LVEF monitoring criteria and the dosing algorithm to avoid drug-induced heart failure/LVEF depression. **Education Protocol (Patient):** Education on reporting symptoms of heart failure or exertional syncope. **Quality Audit Mechanism:** Dual reporting incorporating site-read TTEs for immediate clinical decisions, followed by retrospective central core lab reviews for data integrity.

14. Observation & Follow-Up Schedule

Total Duration: 252 weeks (~5 years) **Onsite Visit Cadence:** Intensive monitoring at Weeks 4, 8, 12, and 24, followed by maintenance visits every 12 weeks (e.g., Weeks 156, 180, etc.). **Remote Monitoring Frequency:** Supplemental symptom tracking as required per site clinical practices. **Checklist Review Interval:** Echocardiograms and NYHA class evaluated at every 12-week clinic visit during the maintenance phase.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				